

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program	NetworkWalks Internship Program — Cybersecurity & Ethical Hacking
Date	17 August 2026
Modules completed	Module 1: Footprinting & Reconnaissance Research (theHarvester) Module 2: Network Scanning Report (Zenmap)
Client/Target	1. microsoft.com (public OSINT — passive, no direct interaction with live infrastructure) 2. My own local lab network (VirtualBox host-only network, 192.168.56.0/24)
Permission secured from client?	Yes — public passive OSINT sources only for Module 1; own isolated lab environment for Module 2
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3–5: In Progress

1. LIABILITY DISCLAIMER

I have performed these activities only on systems and information sources where such activity is permitted: passive, publicly available OSINT sources for the footprinting module, and my own isolated lab network for the scanning module. All materials in this report are for education and research purposes only as part of the NetworkWalks Cybersecurity & Ethical Hacking internship program. Do not use anything from here to break the law. The instructor, the authors, and NetworkWalks are not responsible for what you do with this knowledge. Every action taken is my own responsibility. Misuse of these techniques can lead to criminal charges, heavy fines, loss of employment, and a permanent criminal record. In most countries, unauthorised access to computer systems is a crime even when no damage occurs.

2. INTRODUCTION

This report covers two practical modules completed during Week 2 of my Cybersecurity & Ethical Hacking internship at NetworkWalks. Module 1 covers footprinting and reconnaissance research against the microsoft.com domain using theHarvester, an OSINT aggregation tool. Module 2 covers active network scanning of my own local lab network using Zenmap (the graphical front-end for Nmap). Together, the two modules demonstrate how a penetration tester moves from passive, publicly available information gathering to active host discovery on a live network.

All theHarvester commands were run in Kali Linux 2026.1 (running inside VirtualBox). The Zenmap scan was run from the same Kali Linux environment against an isolated VirtualBox host-only lab network. Every step below includes the exact command used, the result observed, a screenshot as evidence, and a short note on why the finding matters from a security perspective.

3. TOOLS USED

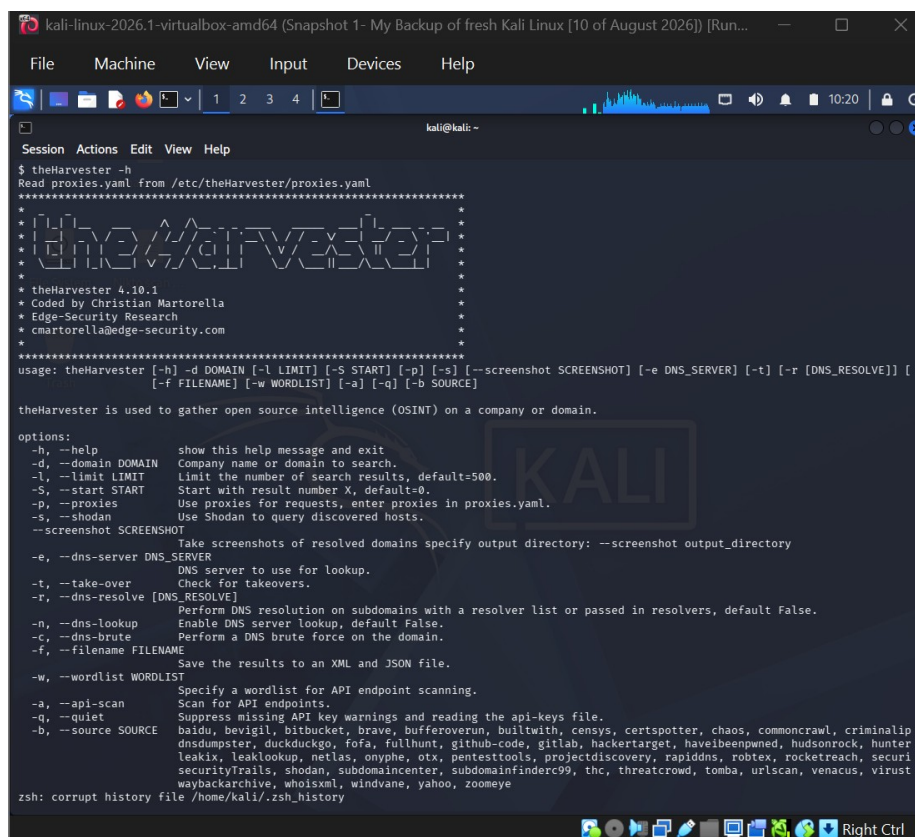
The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux 2026.1 (VirtualBox)	Operating system used for both reconnaissance and scanning activities
theHarvester v4.10.1	OSINT aggregation tool used to passively harvest emails, subdomains/hosts, IP addresses, and ASNs for a target domain
Zenmap (Nmap 7.991 GUI)	Graphical front-end for Nmap, used to perform host discovery on the local lab subnet and identify live hosts with their IP and MAC addresses

4. ACTIVITIES PERFORMED

4.1 Module 1 — Footprinting & Reconnaissance Research (theHarvester)

For the first activity, I performed passive OSINT reconnaissance against the domain microsoft.com using theHarvester. The goal of this phase is to identify publicly available information — email addresses, subdomains, hosts, IP ranges, and Autonomous System Numbers (ASNs) — without directly interacting with the target's live infrastructure. I first reviewed the tool's help menu to confirm the available options and the full list of supported OSINT sources before running any scans.



```
kali@kali: ~$ theHarvester -h
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****
usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-s START] [-p] [-s] [--screenshot SCREENSHOT] [-e DNS_SERVER] [-t] [-r [DNS_RESOLVE]] [-f FILENAME] [-w WORDLIST] [-a] [-q] [-b SOURCE]

theHarvester is used to gather open source intelligence (OSINT) on a company or domain.

options:
  -h, --help            show this help message and exit
  -d, --domain DOMAIN    Company name or domain to search.
  -l, --limit LIMIT      Limit the number of search results, default=500.
  -s, --start START      Start with result number X, default=0.
  -p, --proxies          Use proxies for requests, enter proxies in proxies.yaml.
  -s, --shodan           Use Shodan to query discovered hosts.
  --screenshot SCREENSHOT Take screenshots of resolved domains specify output directory: --screenshot output_directory
  -e, --dns-server DNS_SERVER DNS server to use for lookup.
  -t, --take-over        Check for takeovers.
  -r, --dns-resolve [DNS_RESOLVE] Perform DNS resolution on subdomains with a resolver list or passed in resolvers, default False.
  -n, --dns-lookup       Enable DNS server lookup, default False.
  -c, --dns-brute        Perform a DNS brute force on the domain.
  -f, --filename FILENAME Save the results to an XML and JSON file.
  -w, --wordlist WORDLIST Specify a wordlist for API endpoint scanning.
  -a, --api-scan         Scan for API endpoints.
  -q, --quiet            Suppress missing API key warnings and reading the api-keys file.
  -b, --source SOURCE    baidu, bevigil, bitbucket, brave, bufferoverun, builtwith, censys, certspotter, chaos, commoncrawl, criminalip,
                        dnsdumpster, duckduckgo, fofa, fullhunt, github-code, gitlab, hackertarget, haveibeenpwned, hudsonrock, hunter,
                        leakiv, leaklookup, netlas, onyphe, otx, pentesttools, projectdiscovery, rapiddns, robtex, rocketreach, securi,
                        securitytrails, shodan, subdomaincenter, subdomainfinder99, thc, threatcrowd, tomba, urlscan, venacius, virust,
                        waybackarchive, whoisxml, windvane, yahoo, zoomeye

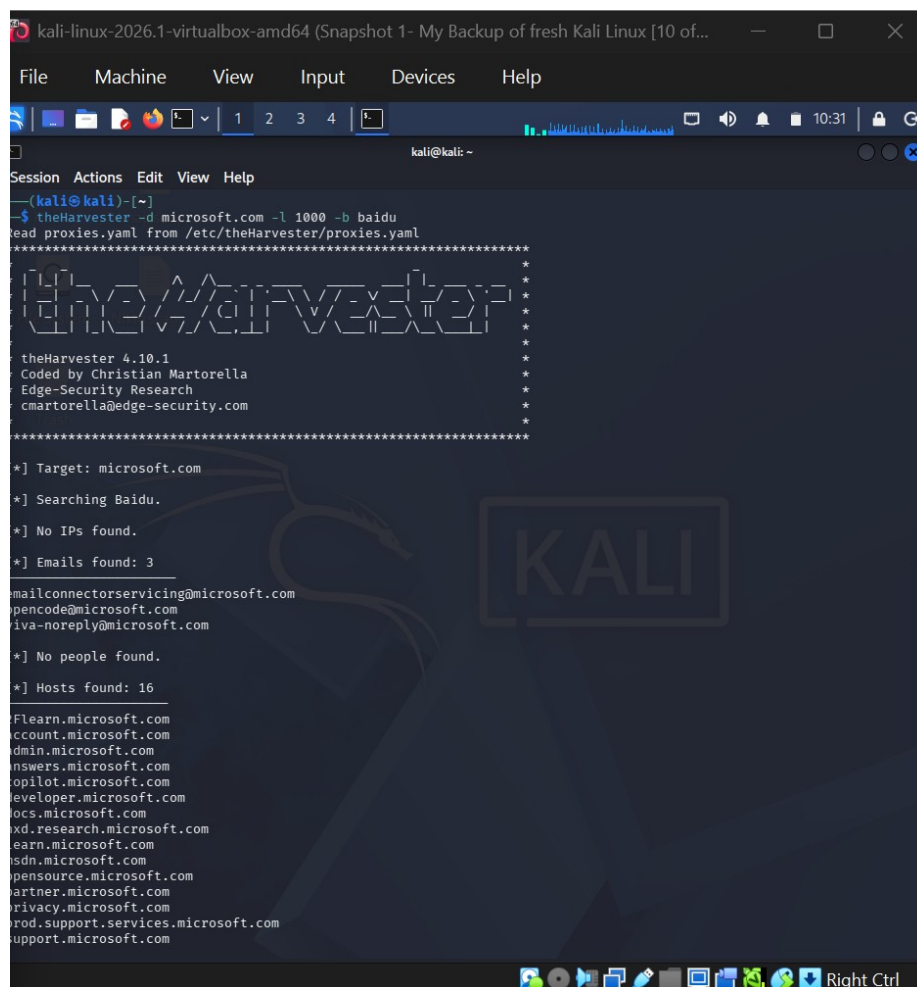
zsh: corrupt history file /home/kali/.zsh_history
```

Figure 1 — theHarvester help/usage output (theharvester -h), showing available flags and the full list of supported OSINT sources.

Scan 1: Baidu source only

Command used: `theHarvester -d microsoft.com -l 1000 -b baidu`

This scan queried the Baidu search engine only, with a result limit of 1000. The results returned 3 email addresses (emailconnectorservicing@microsoft.com, opencode@microsoft.com, viva-noreply@microsoft.com) and 16 hosts, including account.microsoft.com, admin.microsoft.com, copilot.microsoft.com, developer.microsoft.com, learn.microsoft.com, and support.microsoft.com. No IP addresses were returned, reflecting the limited scope of a single-engine passive search.



```
kali@kali: ~  
--(kali@kali)-[~]  
└─$ theHarvester -d microsoft.com -l 1000 -b baidu  
Read proxies.yaml from /etc/theHarvester/proxies.yaml  
*****  
theHarvester 4.10.1  
Coded by Christian Martorella  
Edge-Security Research  
cmartorella@edge-security.com  
*****  
*] Target: microsoft.com  
*] Searching Baidu.  
*] No IPs found.  
*] Emails found: 3  
emailconnector.servicing@microsoft.com  
pencode@microsoft.com  
iva-noreply@microsoft.com  
*] No people found.  
*] Hosts found: 16  
Flearn.microsoft.com  
account.microsoft.com  
admin.microsoft.com  
answers.microsoft.com  
opilot.microsoft.com  
developer.microsoft.com  
docs.microsoft.com  
rd.research.microsoft.com  
learn.microsoft.com  
msdn.microsoft.com  
open-source.microsoft.com  
partner.microsoft.com  
privacy.microsoft.com  
prod.support.services.microsoft.com  
support.microsoft.com
```

Figure 2 — theHarvester output for the Baidu-only scan against microsoft.com, showing 3 harvested emails and 16 discovered hosts.

Scan 2: All configured sources

Command used: `theHarvester -d microsoft.com -l 50 -b all`

This scan attempted to query every source built into theHarvester, including premium/API-gated services such as Shodan, VirusTotal, Censys, Hunter, SecurityTrails, and BuiltWith. Since no API keys were configured, each of these returned a 'Missing API key' message and was automatically skipped, without stopping the overall scan. The scan still succeeded using free, unauthenticated sources such as Baidu, Certspotter, DuckDuckGo, Commoncrawl, Hackertarget, Hudson Rock, CRTsh (Certificate Transparency logs), Otx, Rapiddns, Robtex, Urlscan, and the Wayback Machine archive.

```
kali-linux-2026.1-virtualbox-amd64 (Snapshot 1- My Backup of fresh Kali Linux [10 of...])
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ theHarvester -d microsoft.com -l 50 -b all
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*                                     *
*  THE HARVESTER  *
*                                     *
* theHarvester 4.10.1 *
* Coded by Christian Martorella *
* Edge-Security Research *
* cmartorella@edge-security.com *
*                                     *
*****

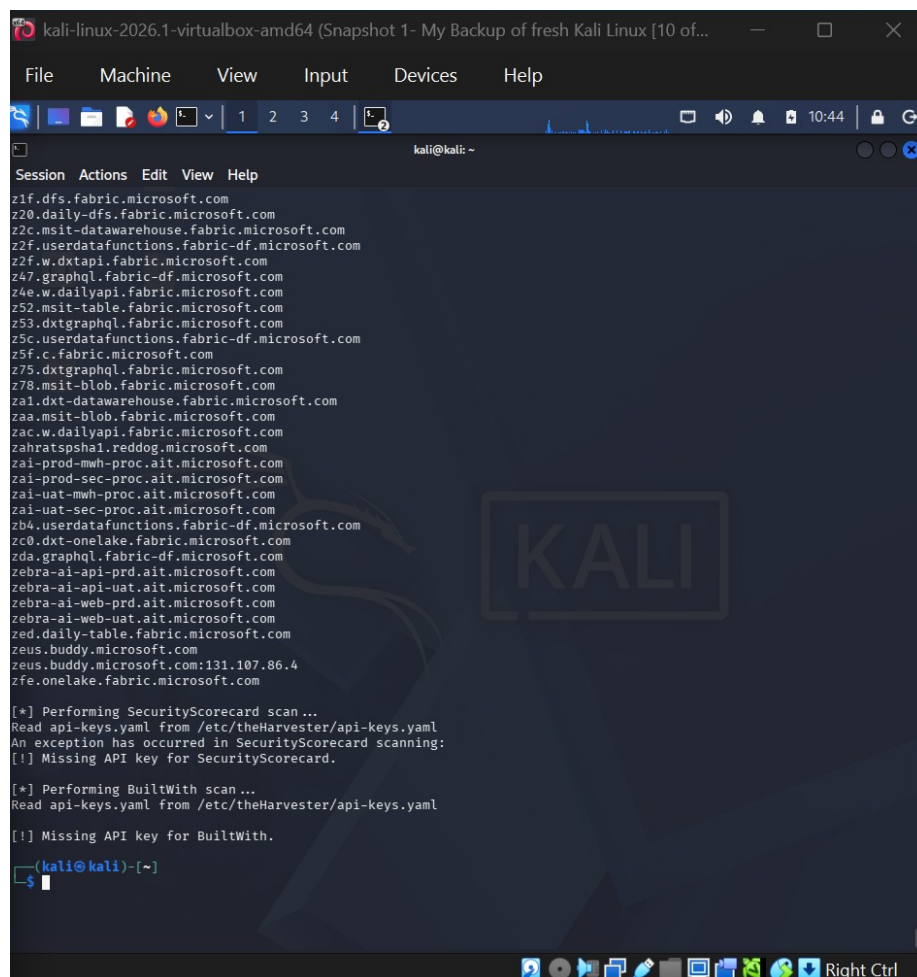
[*] Target: microsoft.com

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverun search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bufferoverun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: 'microsoft.com'
A Missing Key Error occurred in builtwith:
[!] Missing API key for BuiltWith.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: 'microsoft.com'
Error Message:
[!] Missing API key for Brave Search.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Censys API key is missing or invalid:
[!] Missing API key for Censys ID and/or Secret.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing key error occurred in criminalip:
[!] Missing API key for criminalip.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
```

Figure 3 — Start of the all-sources scan, showing theHarvester reading api-keys.yaml and reporting missing-key errors for unconfigured premium sources.

The results identified 3 emails (dotnet-docker-bot@microsoft.com, opencode@microsoft.com, secure@microsoft.com), 4 ASNs (AS8075, AS13335, AS40034, AS133618), 119 IP addresses spanning Microsoft's cloud/edge ranges, and 9,951 subdomains/hosts — including a large number of pre-production-labelled hosts (using -ppe, -uat, -dev, and -test suffixes) and Fabric data-platform services.



```
kali-linux-2026.1-virtualbox-amd64 (Snapshot 1- My Backup of fresh Kali Linux [10 of...])
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

z1f.dfs.fabric.microsoft.com
z20.daily-dfs.fabric.microsoft.com
z2c.msit-datawarehouse.fabric.microsoft.com
z2f.userdatafunctions.fabric-df.microsoft.com
z2f.w.dxtapi.fabric.microsoft.com
z47.graphql.fabric-df.microsoft.com
z4e.w.dailyapi.fabric.microsoft.com
z52.msit-table.fabric.microsoft.com
z53.dxtgraphql.fabric.microsoft.com
z5c.userdatafunctions.fabric-df.microsoft.com
z5f.c.fabric.microsoft.com
z75.dxtgraphql.fabric.microsoft.com
z78.msit-blob.fabric.microsoft.com
za1.dxt-datawarehouse.fabric.microsoft.com
zaa.msit-blob.fabric.microsoft.com
zac.w.dailyapi.fabric.microsoft.com
zahratspsahl.reddog.microsoft.com
zai-prod-mwh-proc.ait.microsoft.com
zai-prod-sec-proc.ait.microsoft.com
zai-uat-mwh-proc.ait.microsoft.com
zai-uat-sec-proc.ait.microsoft.com
zb4.userdatafunctions.fabric-df.microsoft.com
zc0.dxt-onelake.fabric.microsoft.com
zda.graphql.fabric-df.microsoft.com
zebra-ai-api-prd.ait.microsoft.com
zebra-ai-api-uat.ait.microsoft.com
zebra-ai-web-prd.ait.microsoft.com
zebra-ai-web-uat.ait.microsoft.com
zed.daily-table.fabric.microsoft.com
zeus.buddy.microsoft.com
zeus.buddy.microsoft.com:131.107.86.4
zfe.onelake.fabric.microsoft.com

[+] Performing SecurityScorecard scan ...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
An exception has occurred in SecurityScorecard scanning:
[!] Missing API key for SecurityScorecard.

[*] Performing BuiltWith scan ...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for BuiltWith.

(kali@kali)-[~]
$
```

Figure 4 — Tail end of the all-sources scan, showing the final batch of discovered *.fabric.microsoft.com hosts, followed by the SecurityScorecard and BuiltWith scan attempts, both skipped for missing API keys.

A notable observation was a cluster of misspelling variants resolving under the same domain — `account.microsoft.com`, `accoun.microsoft.com`, `accuont.microsoft.com`, and `acount.microsoft.com` — which may indicate either defensive typo registrations or legacy DNS records worth flagging for review.

4.2 Module 2 — Network Scanning Report (Zenmap)

For the second activity, I used Zenmap to perform active host discovery on my own isolated VirtualBox lab network. The task required identifying the lab subnet, discovering live hosts, and recording their IP and MAC addresses before any further port or service enumeration would be carried out.

Command used: `nmap -sn 192.168.56.0/24`

The `-sn` flag instructs Nmap to perform host discovery only (a 'ping scan'), sending probes to every address in the target range without proceeding to port scanning. This is the appropriate first step before running deeper, noisier scans against individual targets.

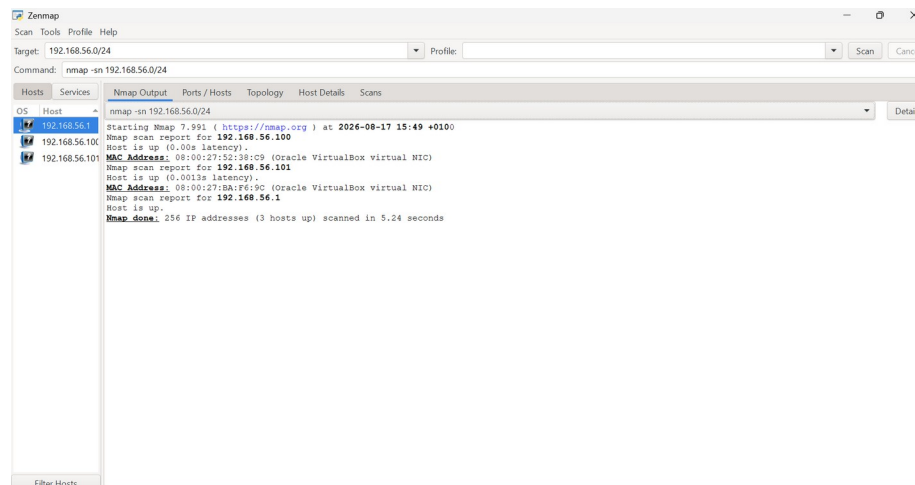


Figure 5 — Zenmap GUI showing the completed host-discovery scan (`nmap -sn 192.168.56.0/24`) against the lab network, with 3 of 256 addresses responding as live.

The scan completed in 5.24 seconds and identified 3 live hosts out of 256 scanned addresses:

- 192.168.56.1 — Up — MAC 0A:00:27:00:00:14 (host machine / VirtualBox host-only adapter)
- 192.168.56.100 — Up — MAC 08:00:27:52:38:C9 (Oracle VirtualBox virtual NIC — guest VM)
- 192.168.56.101 — Up — MAC 08:00:27:BA:F6:9C (Oracle VirtualBox virtual NIC — guest VM)

The MAC address vendor identification (Oracle VirtualBox virtual NIC) confirms that the two remote hosts are virtualized targets, which is expected and appropriate for a controlled lab exercise. No port or service information was captured in this pass, since `-sn` deliberately skips the port-scanning phase — this scan is documented as discovery-only, with port and service enumeration planned as a follow-up activity.

5. RISK ANALYSIS / IMPACT

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Large external attack surface exposed	theHarvester identified 9,951 subdomains and 119 IP addresses for microsoft.com	A large footprint increases the number of potential entry points an attacker could investigate further	● Medium
2	Pre-production/ staging hosts publicly resolvable	Numerous hosts used - dev, -test, -uat, and -ppe naming suffixes	Staging environments are statistically more likely to be less hardened than production systems	● Medium
3	Service/ automation email addresses exposed	theHarvester identified 3 emails per scan (e.g. <code>opencode@microsoft.com</code> , <code>secure@microsoft.com</code>)	Exposed addresses could be used as a starting point for phishing-pretext development	● Low
4	Legacy/typo DNS records present	Misspelling variants (<code>acctont.</code> , <code>accoun.</code> , <code>account.</code> , <code>accuont.</code>) resolve under microsoft.com	May indicate legacy or forgotten DNS records that should be reviewed and possibly removed	● Low
5	Incomplete OSINT coverage without API keys	Premium sources (Shodan, VirusTotal, Censys, SecurityTrails, BuiltWith) failed due to missing API keys	The true external footprint is likely larger than what free-tier sources revealed	● Low
6	Multiple live hosts visible on local lab network	Zenmap identified 3 live hosts on the 192.168.56.0/24 lab subnet	On a real network, unknown or unauthorized devices could be present and should be investigated	● Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. Both modules involved information gathering and host discovery only — no exploitation or vulnerability validation was performed. The presence of a subdomain, IP address, or DNS record does not by itself mean that a system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. RECOMMENDATIONS

Based on the observations from these activities, I recommend the following security improvements:

- 1. Review the external subdomain footprint regularly** — Organizations should periodically enumerate their own external subdomains to understand what is publicly discoverable and remove or secure anything unnecessary.
- 2. Harden or restrict access to pre-production environments** — Staging, UAT, dev, and PPE-labelled hosts should not be reachable from the public internet unless required, and should be held to the same security standard as production.
- 3. Review publicly exposed contact/service email addresses** — Ensure that only intended addresses are exposed and that automated/service accounts are monitored for abuse.
- 4. Clean up legacy and typo DNS records** — DNS zones should be audited periodically to remove stale, forgotten, or unnecessary records.
- 5. Perform OSINT reconnaissance with full source coverage** — Where authorized, use premium OSINT sources (Shodan, VirusTotal, SecurityTrails, Censys) for a more complete view of the external footprint.
- 6. Perform regular internal network discovery** — Organizations should periodically scan their own networks to maintain an accurate inventory of active devices.
- 7. Investigate unknown devices** — Any unexpected device discovered during network scanning should be investigated and verified against the asset inventory.
- 8. Proceed to port and service enumeration** — Follow up the host-discovery scan with a full TCP/UDP port and service scan against the confirmed live lab hosts.
- 9. Perform security testing with authorization** — Reconnaissance and scanning should only be performed against systems, domains, and networks where appropriate authorization has been secured.

7. CONCLUSION

During Week 2 of my Cybersecurity & Ethical Hacking internship at NetworkWalks, I completed two practical modules covering footprinting, reconnaissance research, and network scanning.

In Module 1, I used theHarvester to passively collect OSINT information about the microsoft.com domain across two scans — a single-source (Baidu) run and an all-sources run. I learned how the tool aggregates data from search engines, certificate transparency logs, DNS-based fallback methods, and other free/unauthenticated sources, and how it gracefully skips premium sources when API keys are not configured. The exercise showed me how large an organization's external footprint can be, and how naming conventions (like -dev, -test, -uat) can reveal potentially weaker points in that footprint.

In Module 2, I used Zenmap to perform host discovery on my own isolated lab network, identifying the live hosts and their IP/MAC addresses before any deeper scanning was attempted. This confirmed the expected lab topology and gave me hands-on experience with the first step of active network reconnaissance.

These exercises showed me that information gathering is a critical part of cybersecurity. Even before attempting to interact further with a system, a security professional can learn a significant amount about a target environment by carefully analyzing publicly available information and network responses. I also learned that technical findings should be documented clearly — a good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab under the NetworkWalks internship program.

-End-

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Project Information

Program Name: Cybersecurity & Ethical Hacking Internship at NetworkWalks | Week: 02 | Modules: theHarvester (Footprinting) + Zenmap (Network Scanning)